

NEUROX

3D NEUROLOGICAL BRAIN SEGMENTATION

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SCHOOL: SCOPE

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PROBLEM STATEMENT

The core problem addressed by this project is the lack of a unified, high-efficiency system for simultaneously analyzing multiple neurological conditions from a single MRI scan. Current clinical workflows face several challenges:

Siloed Diagnostic Tools:

Existing AI models are typically designed for a single task (e.g., only tumor segmentation), requiring clinicians to run multiple separate systems for a comprehensive patient assessment.

Inconsistent Data & Quality:

Using separate models can lead to inconsistent data quality and negative transfer, making it difficult to achieve a unified view of a patient's neurological health.



Clinical Uncertainty:

Many existing AI assistants provide "black-box" predictions without quantifying diagnostic uncertainty, which is critical for medical professionals making treatment decisions.

Complex Coexistence:

Critical conditions like brain tumors, strokes, and Alzheimer's disease can coexist, yet they are rarely analyzed together in a single, high-performance deep learning pipeline.



MOTIVATION

The motivation for developing NeuroX stems from the need to provide a "clinically-viable AI diagnostic assistant" that can streamline complex neurological evaluations:

Unified Multi-Disease Diagnostics:

The project was driven by the need to replace siloed tools with a single system capable of detecting Brain Tumors, Stroke, and Alzheimer's from one MRI scan, significantly reducing clinical analysis time.

High-Fidelity Precision and Visualization:

A primary motivation was to assist medical professionals in accurately locating and quantifying pathologies through the generation of precise 3D brain visualizations and high-resolution 2D clinical heatmaps.

Clinical Trust via Uncertainty Quantification:

To bridge the gap between AI predictions and clinical decision-making, the system incorporates Monte Carlo Dropout and heteroscedastic uncertainty to flag complex cases for immediate expert review.

Scientific Rigor and Automated Reporting:

The system ensures real-world utility by meeting publication-grade statistical standards through Nested Cross-Validation and simplifies physician workflows with an automated, AI-driven clinical reporting engine.



OBJECTIVES

- Develop a unified AI-based system for analysing brain MRI scans.
- Detect multiple neurological disorders including brain tumor, stroke, and Alzheimer's disease.
- Apply MRI preprocessing techniques such as normalization and skull stripping for standardized input.
- Extract deep volumetric features using 3D CNN and transformer-based architecture.
- Perform accurate lesion segmentation for abnormal brain region localization.
- Generate confidence-aware predictions and automated diagnostic reports.





LITERATURE REVIEW

Paper / Method	Techniques Used	Key Findings	Limitation
Attention U-Net	U-Net with attention gates for medical image segmentation	Improved lesion localization and segmentation accuracy	Focused only on segmentation, no classification support
Transformer-Based MRI Diagnosis	CNN + Transformer self-attention mechanism	Captures global spatial relationships effectively	High computational complexity
3D CNN for Brain MRI Classification	Volumetric 3D convolution layers	Strong feature extraction from MRI volumes	Limited interpretability and no reporting system
Automated Brain Tumor Segmentation Models	CNN-based segmentation methods	Accurate tumor boundary detection	Works only for tumor-specific diagnosis



LITERATURE REVIEW

Paper / Method	Techniques Used	Key Findings	Limitation
Deep Learning-Based Stroke Lesion Segmentation	CNN-based lesion segmentation	Improved stroke lesion detection in MRI	Sensitive to dataset imbalance
MRI-Based Alzheimer Detection Models	Deep CNN classification	Detects structural degeneration patterns	Limited generalization across datasets
Hybrid CNN-Transformer Medical Models	CNN + Transformer hybrid architecture	Better local and global feature learning	Computationally intensive
Our Proposed System (NeuroX)	3D CNN + Transformer + Attention Decoder	Unified multi-disease MRI diagnosis	Requires larger clinical validation

- **Limitations in Existing Systems**

- Most existing systems focus on single neurological disease detection only.
- Classification and segmentation are usually developed as separate modules.
- Existing models often lack confidence estimation for prediction reliability.
- Automated clinical report generation is rarely integrated.
- Many systems are difficult to generalize across MRI datasets from different sources.

- **Technical Gap**

- Limited use of unified architectures combining 3D CNN and transformer learning.
- Lack of practical frameworks supporting tumor, stroke, and Alzheimer detection together.

- **Gap Addressed by NeuroX**

- NeuroX proposes a unified pipeline integrating multi-disease detection, segmentation, confidence estimation, visualization, and report generation.



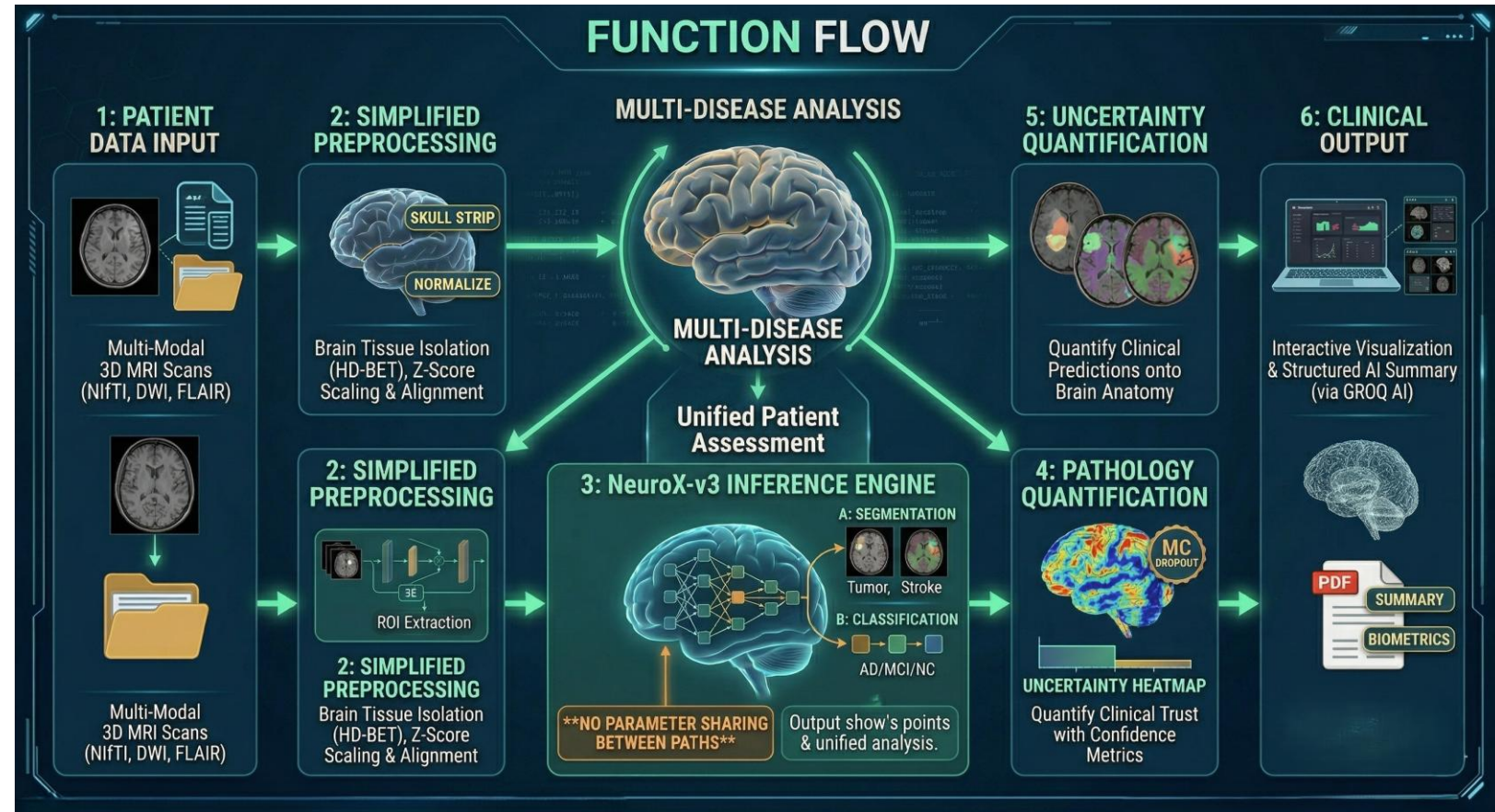
PROPOSED SYSTEM

- **Overview of the Solution**

- NeuroX is a unified AI-based medical imaging framework designed to analyze brain MRI scans for detecting multiple neurological disorders within a single diagnostic pipeline.

- **The proposed system performs:**

- Brain tumor detection
- Stroke lesion detection
- Alzheimer pattern identification
- Lesion segmentation
- Confidence-aware prediction
- Automated report generation





PROPOSED SYSTEM

INNOVATION AND ENGINEERING CONTRIBUTION

- **Innovation / Novelty**

- Unified diagnosis for tumor, stroke, and Alzheimer's disease
- Hybrid 3D CNN + Transformer architecture
- Attention-guided decoder for segmentation
- Confidence estimation using uncertainty analysis

- **Engineering Contribution**

- Modular medical inference pipeline
- Automated PDF reporting system
- Streamlit-based clinical interface
- Integrated 3D lesion visualization

- **Core Contribution**

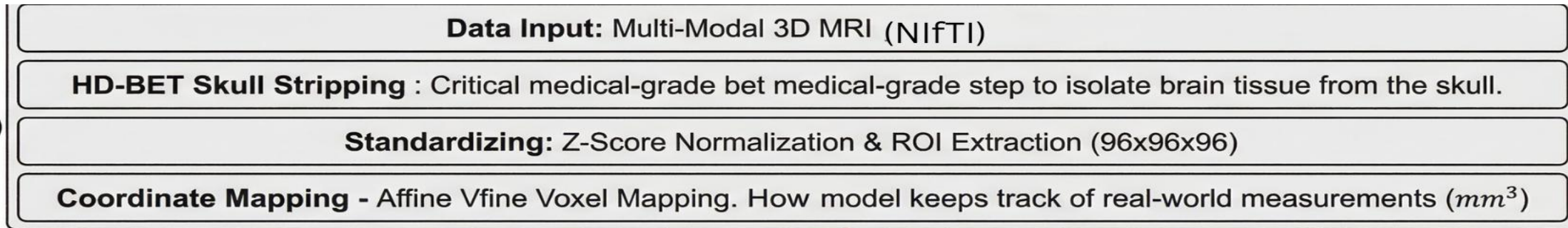
- NeuroX combines multiple medical AI tasks into one deployable framework.



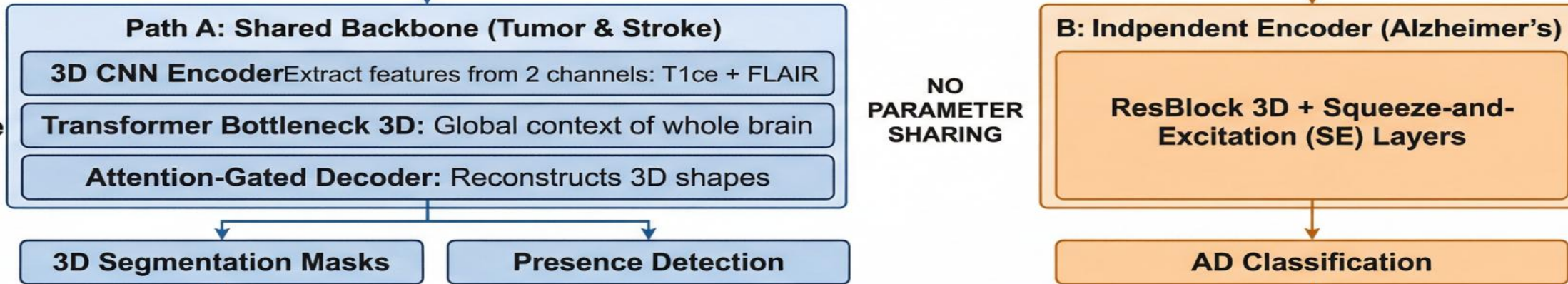


SYSTEM ARCHITECTURE BLOCK DIAGRAM

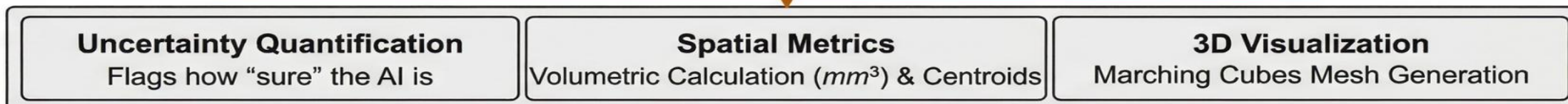
Stage 1: Preprocessing (The Foundation)



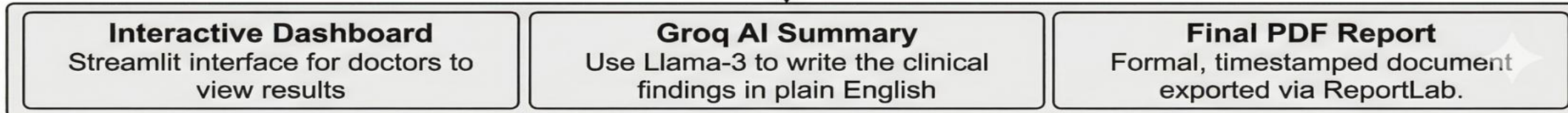
Stage 2: NeuroX-v3 Inference Engine (The Core)



Stage 3: Post-Analysis & Uncertainty (The Rigor)



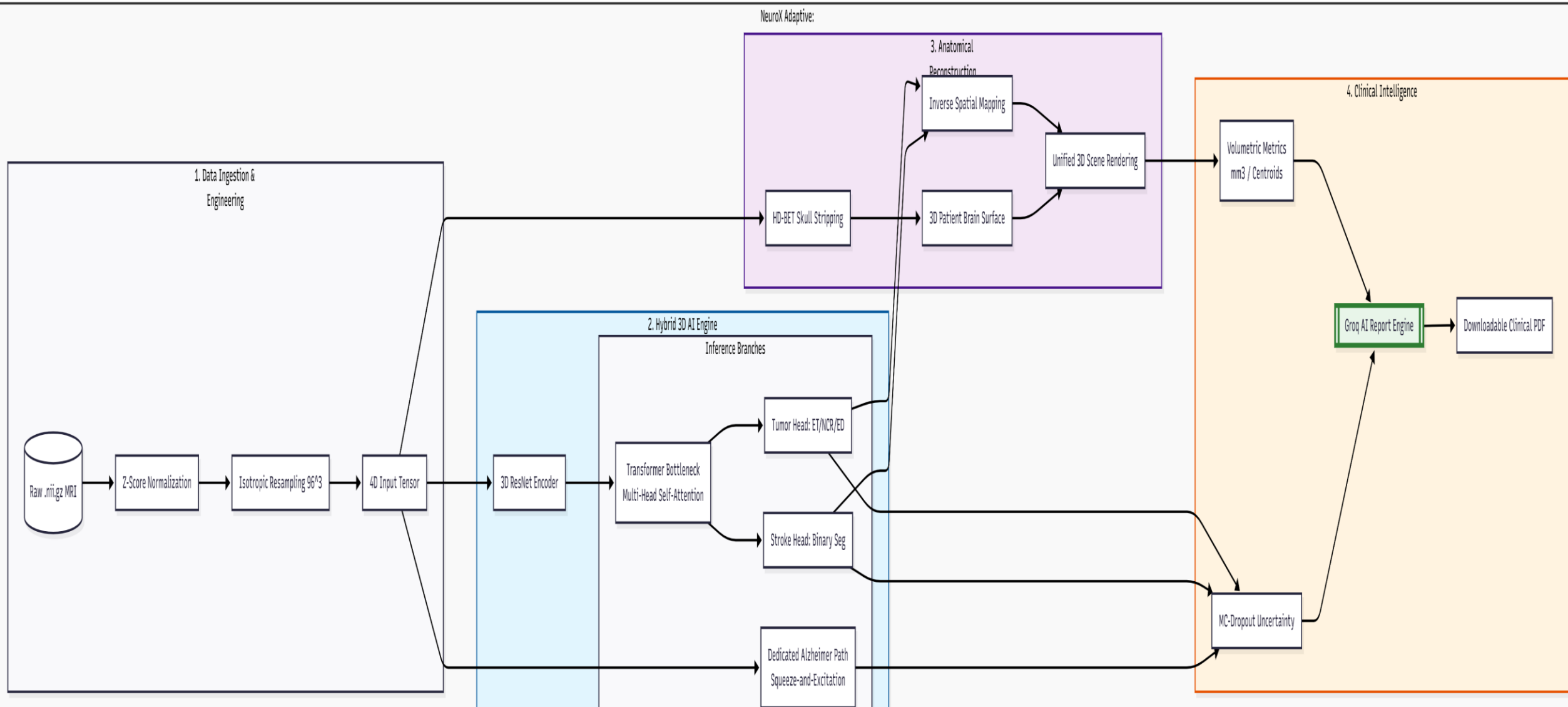
Stage 4: Clinical Output (The Result)

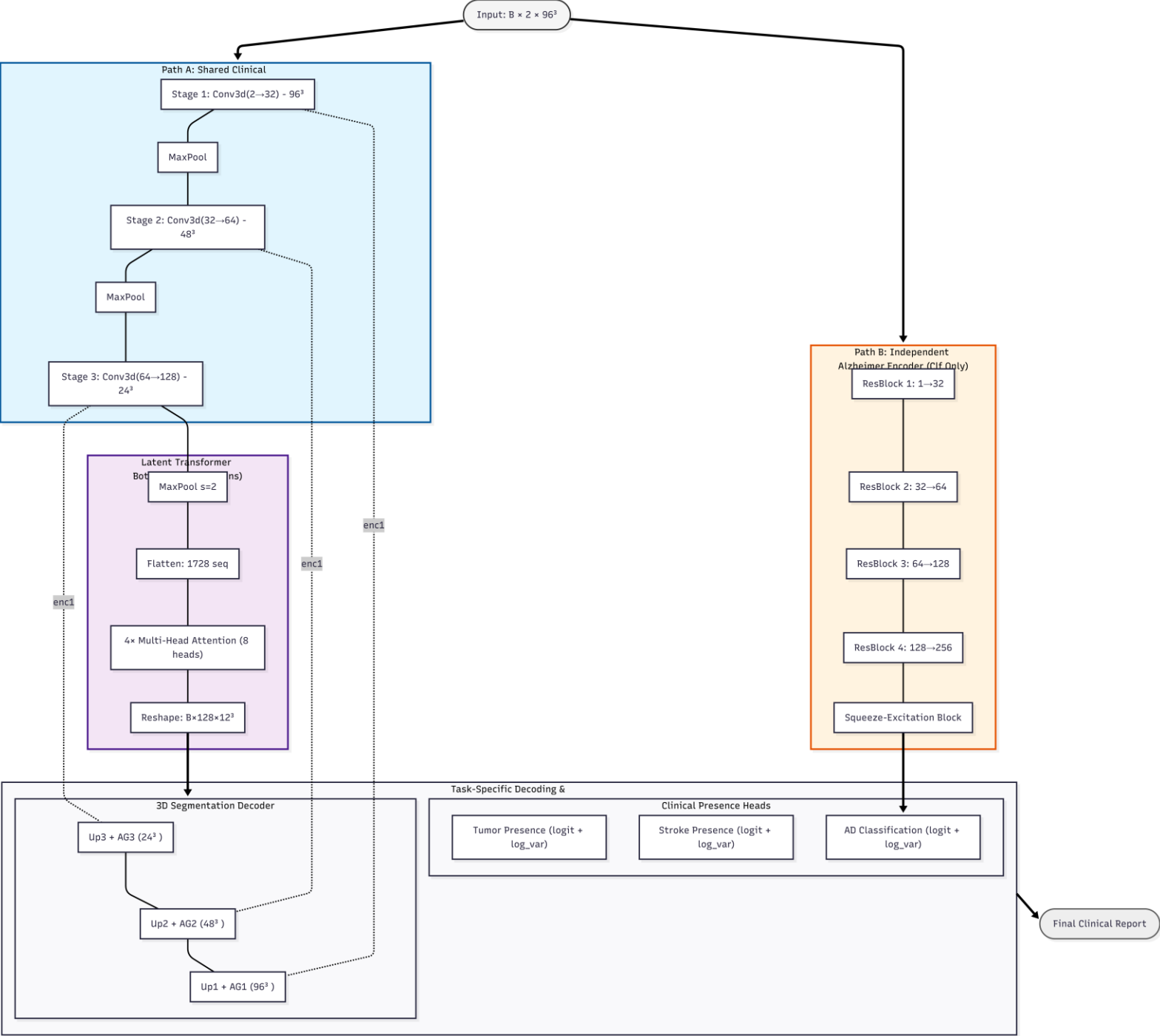




SYSTEM ARCHITECTURE

ARCHITECTURE DIAGRAM





SYSTEM ARCHITECTURE

DATA FLOW DIAGRAM



METHODOLOGY

ALGORITHM

1. Preprocessing & Input Routing

The algorithm begins by standardizing the input MRI to ensure consistency across different scanners:

- **Normalization:** Voxel intensities are normalized using Z-score techniques and clipped to remove outliers.
- **Spatial Alignment:** The 3D volume is resampled to a uniform $96 \times 96 \times 96$ Region of Interest (ROI).
- **Channel Packing:** For tumor and stroke tasks, the system uses a dual-channel input (typically T1ce and FLAIR sequences) to provide the model with better contrast for focal lesions.



METHODOLOGY

ALGORITHM

2. The Core Architecture (NeuroX-v3)

The system utilizes an asymmetric dual-path engine to process different types of brain pathologies:

Path A: Shared Clinical Backbone (Tumor & Stroke)

- **Encoder:** A 3D Convolutional Neural Network (CNN) with Instance Normalization extracts spatial features.
- **Transformer Bottleneck:** A Latent Transformer models global spatial dependencies, helping the model understand the anatomical context of detected lesions.
- **Segmentation Decoders:** An **Attention-Gated U-Net** structure is used to generate the 3D masks. Attention gates suppress irrelevant features, focusing the model on the precise boundaries of the tumor or stroke.

Path B: Independent Alzheimer's Path

- **Dedicated Encoder:** Alzheimer's detection uses a separate **3D Residual CNN (ResBlock3D)** with Squeeze-and-Excitation (SE) blocks.
- **Global Analysis:** Unlike the focal lesion path, this branch analyzes diffuse patterns of cortical atrophy to classify the presence of Alzheimer's.



METHODOLOGY ALGORITHM

3. Detection & Uncertainty Algorithm

For all three conditions, the algorithm employs a probabilistic approach:

- **Presence Heads:** Fully connected layers output a probability score for each disease.
- **Uncertainty Quantification:** The system uses **Monte Carlo (MC)** **Dropout** and a dedicated **Log-Variance head** to calculate epistemic and aleatoric uncertainty. This provides a confidence level for every detection, identifying cases that may require closer expert review.



METHODOLOGY

ALGORITHM

4. 3D Mesh Generation & Clinical Mapping

To produce the final 3D output, the algorithm executes several geometric steps:

- **Marching Cubes Algorithm:** This is applied to the tumor and stroke segmentation masks to create a 3D surface mesh.
- **Inverse Affine Transformation:** The results are mapped back from the 96^3 ROI to the original patient space using the MRI's **NIfTI Affine Matrix**. This ensures that calculated metrics, such as lesion volume (mm^3) and centroid coordinates (World RAS), are anatomically accurate for the clinical report.

5. Training Curriculum

The model was optimized using a **48-epoch Phase-Aware Curriculum**:

- **Phase 1:** Focuses on warming up the Alzheimer's path.
- **Phase 2:** Initializes the segmentation decoders for tumor and stroke.
- **Phase 3:** Joint training where all paths are optimized together to reach a final calibrated state.



TOOLS & TECHNOLOGIES USED

Visualization & Deployment

- **Streamlit** – Clinical web interface
- **Plotly** – Interactive 3D visualization
- **Matplotlib** – Slice heatmaps
- **Trimesh** – STL mesh export

Evaluation & Reporting

- **Scikit-learn** – Metrics & validation
- **NumPy / Pandas** – Data handling
- **ReportLab** – PDF report generation
- **Groq AI (Llama-3)** – AI-assisted neuroradiology reporting

Core Development

- **Python 3.8+** – Main programming language
- **PyTorch 2.0+** – Deep learning framework
- **CUDA 11.8+** – GPU acceleration
- **python-dotenv** – Secure environment configuration

Medical Imaging & Processing

- **NiBabel** – NIfTI MRI file handling
- **HD-BET** – Brain extraction
- **Scikit-image** – Image processing & 3D reconstruction
- **SciPy** – Morphological operations

AI / Deep Learning Models

- **3D CNN Encoder** – Feature extraction
- **Transformer Bottleneck 3D** – Global context learning
- **Attention-Gated U-Net** – Tumor & stroke segmentation
- **ResBlock3D + SE Layers** – Alzheimer MRI classification



IMPLEMENTATION DETAILS

Modules Developed

- **Preprocessing Module:** MRI loading, normalization, padding, resampling ($96 \times 96 \times 96$)
- **Detection Module:** Shared 3D CNN + transformer bottleneck for tumor/stroke detection
- **Alzheimer Module:** Independent residual encoder with uncertainty head
- **Segmentation Module:** Attention U-Net for tumor and stroke lesion masks
- **Visualization Module:** 3D lesion rendering + brain extraction using HD-BET
- **Frontend Module:** Interactive deployment using Streamlit



IMPLEMENTATION DETAILS

Dataset Used

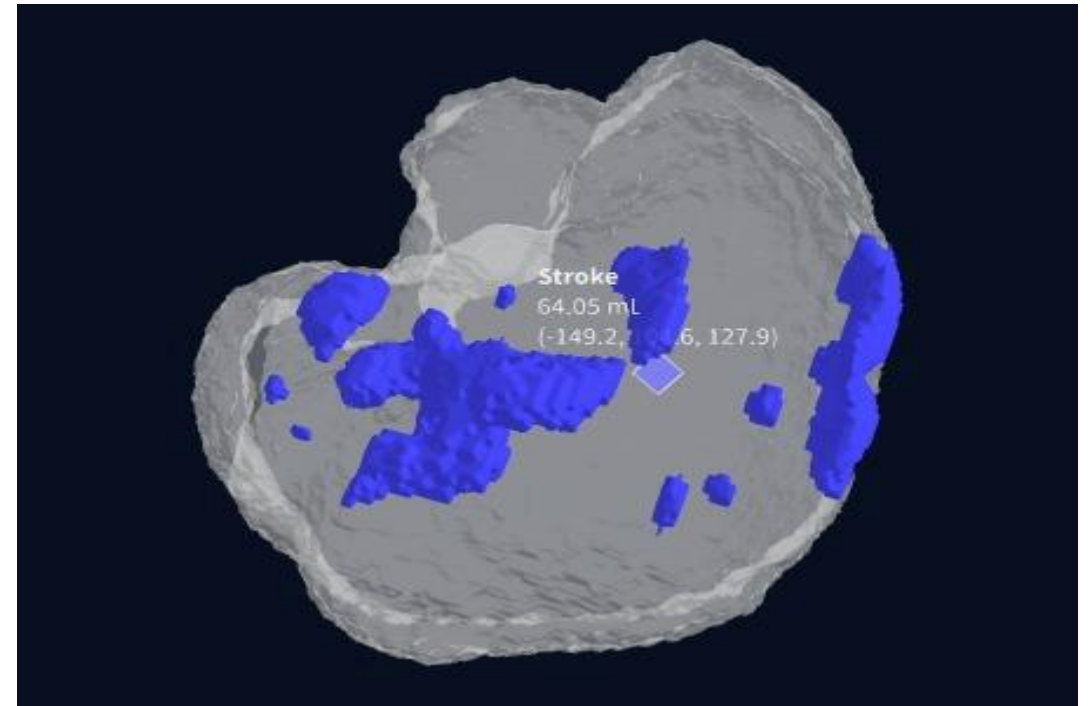
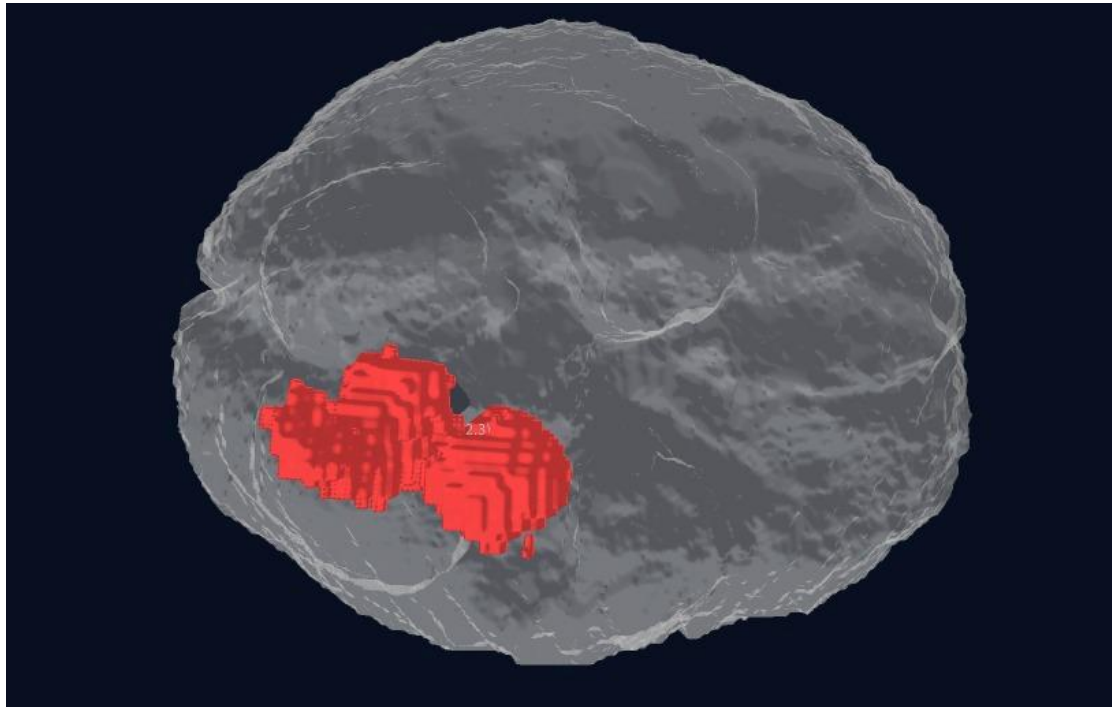
- **Tumor:** BraTS Challenge 2020 + 2021 (**1619 scans**)
- **Stroke:** ISLES Challenge 2022 (**250 scans**)
- **Alzheimer:** Alzheimer's Disease Neuroimaging Initiative (**1208 scans**)

Experimental Setup

- Framework: PyTorch
- Hardware: Tesla T4 GPU
- Training: **48 epochs, 3-phase curriculum learning**
- Batch Size: **1** with gradient accumulation (**4**)
- Loss: **Dice + BCEWithLogitsLoss**
- Calibration: **Temperature Scaling + Monte Carlo Dropout**



RESULTS AND ANALYSIS

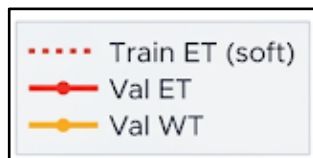
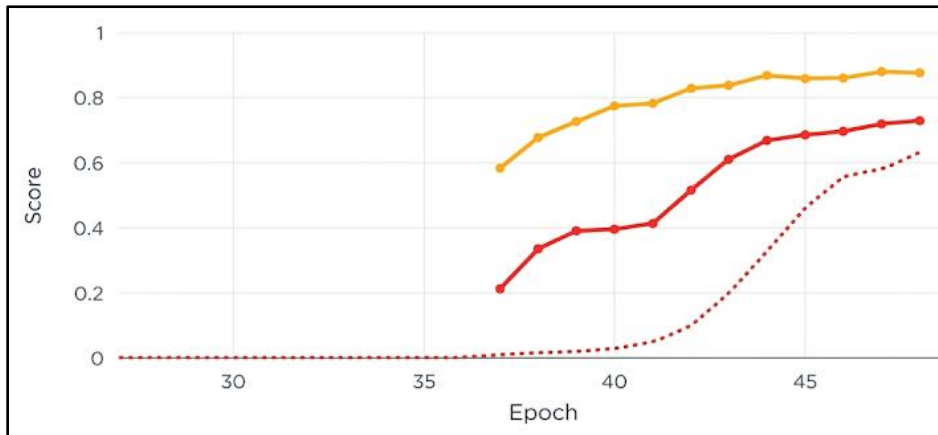




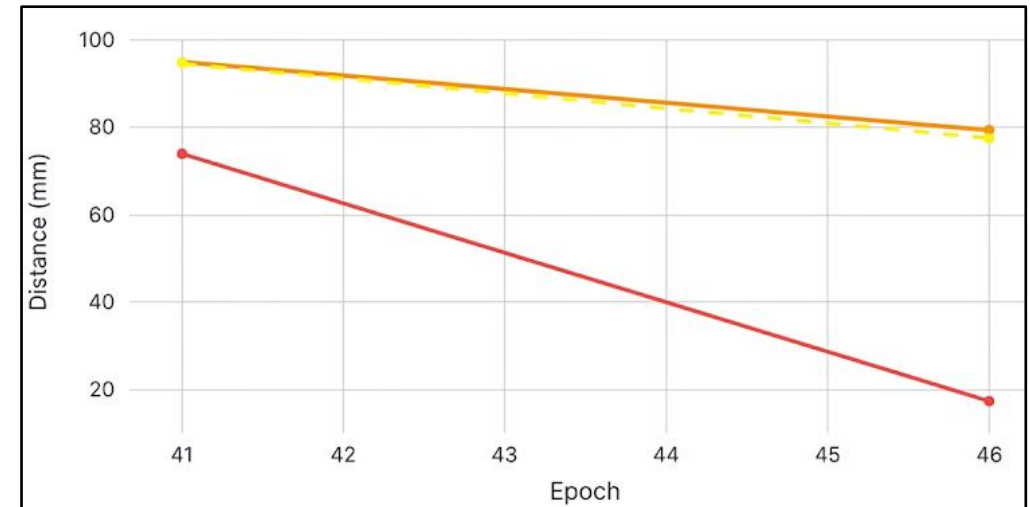
RESULTS AND ANALYSIS

TUMOR SEGMENTATION

Train (soft) vs Val (threshold)



Tumor Val HD95 & ASD (mm) – only HD epochs

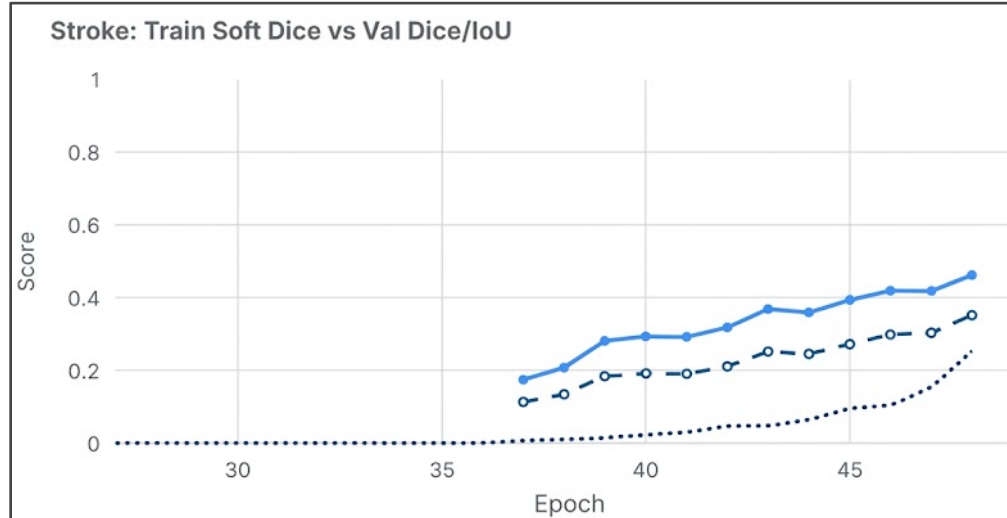




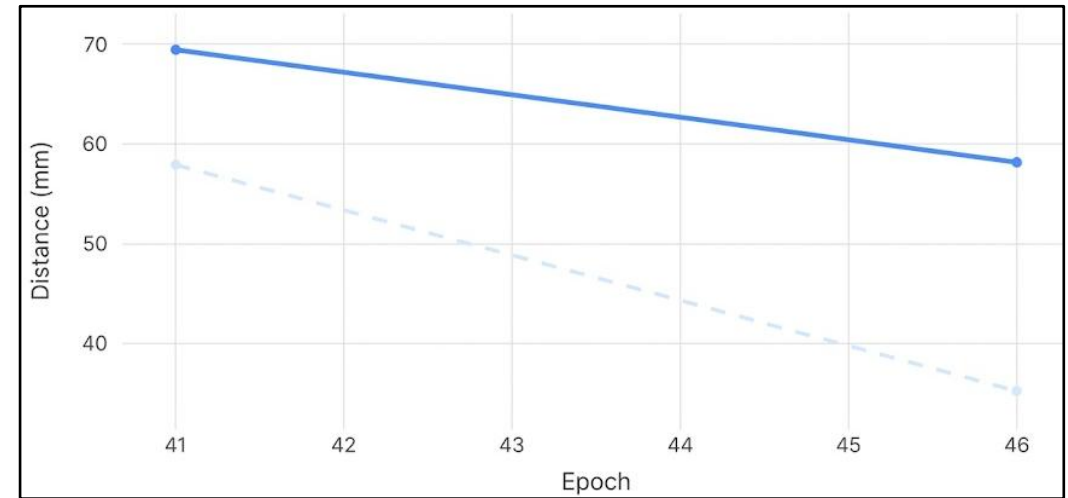
RESULTS AND ANALYSIS

STROKE SEGMENTATION

Train Soft Dice vs Val Dice/IOU



Stroke Val HD95 & ASD (mm) – only HD epochs

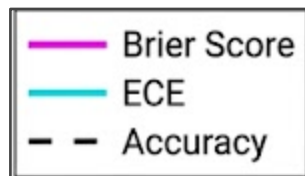
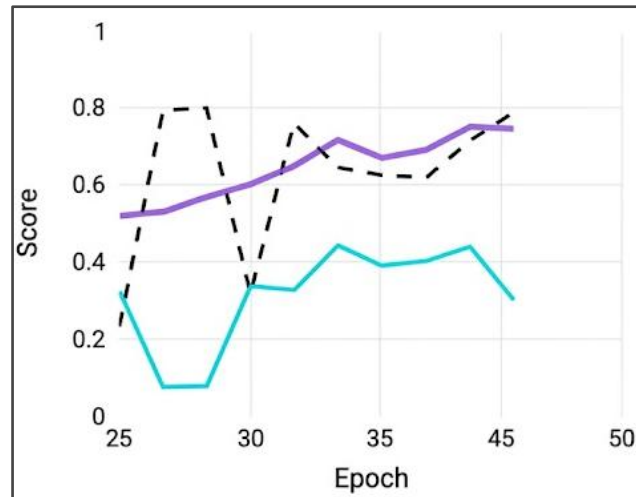




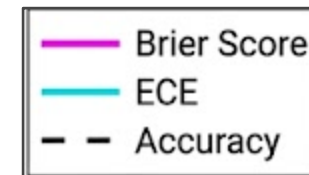
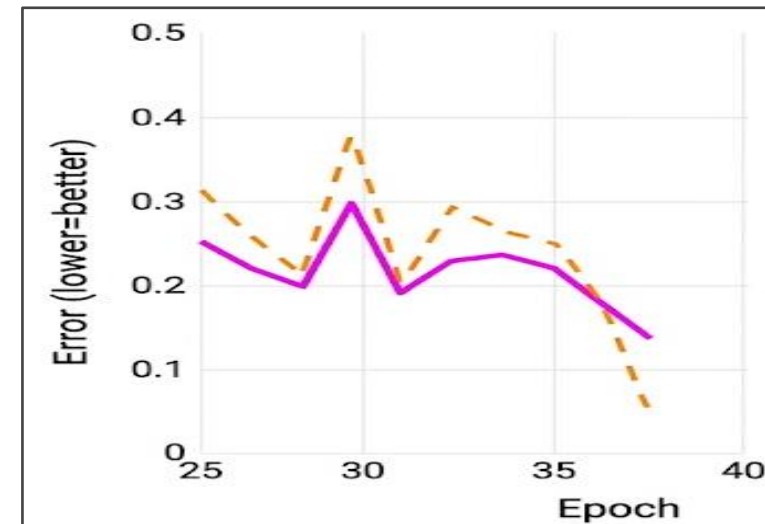
RESULTS AND ANALYSIS

ALZHEIMER SEGMENTATION

Alzheimer Val: AUC / F1 / Accuracy



Alzheimer Val: Calibration (Brier/EUE)





RESULTS AND ANALYSIS

Performance Metrics

Metric	Best Value	Best Epoch	Phase
Tumor ET Dice (val)	0.7637	Ep 26 → held constant	Phase 2 SEG (ep 26)
Tumor TC Dice (val)	0.8322	Ep 26 → held constant	Phase 2 SEG (ep 26)
Tumor WT Dice (val)	0.8902	Ep 26 → held constant	Phase 2 SEG (ep 26)
Stroke Dice (val)	0.5005	Ep 27 onward (plateau)	Phase 3 Joint (ep 27+)
Stroke IoU (val)	0.4051	Ep 27 onward (plateau)	Phase 3 Joint (ep 27+)
Alzheimer Loss (train)	0.2052	Ep 48	Phase 3 Joint (final)
Global Score (best)	2.0385	Ep 34	Phase 3 Joint
Stroke Train Dice	0.6412	Ep 26 (warmup peak)	Phase 2 SEG

COMPARISON: EXISTING VS PROPOSED SYSTEM

Parameter	Existing Systems	Proposed NeuroX System
Disease Coverage	Detects only one disease at a time	Simultaneous detection of tumor, stroke, and Alzheimer's
Model Architecture	Conventional CNN-based single-task models	Shared 3D CNN + transformer + dedicated Alzheimer branch
Segmentation Capability	Limited or disease-specific segmentation	Tumor multi-class + stroke binary segmentation
Uncertainty Estimation	Not available in most systems	Monte Carlo dropout + variance prediction
Visualization	Mostly 2D outputs	Interactive 3D lesion visualization
Brain Extraction	Often manual or basic preprocessing	Automated using HD-BET
Clinical Reporting	Manual interpretation required	Automated PDF clinical report generation
Deployment	Research-only standalone scripts	Interactive deployment using Streamlit
Dataset Handling	Separate models for separate datasets	Unified multi-dataset training framework



APPLICATIONS OF NEUROX

Industry Relevance

- Useful in hospitals and diagnostic centers for assisting MRI-based neurological screening
- Supports radiologists in early detection of tumor, stroke, and Alzheimer's disease
- Can reduce manual workload in multi-scan clinical analysis
- Applicable in medical AI research and decision-support systems
- Suitable for integration with hospital imaging workflows

Real-Time Deployment

- Deployed as interactive web application using **Streamlit**
- Accepts MRI scans in NIfTI format for direct analysis
- Generates disease prediction, segmentation, and 3D visualization in one workflow
- Produces automated PDF clinical reports for rapid review
- Can run on GPU-enabled systems for faster inference

CHALLENGES FACED DURING NEUROX DEVELOPMENT



Major Challenges

- Handling **different MRI modalities** across tumor, stroke, and Alzheimer datasets.
- Managing **large 3D MRI volumes**, which increased memory usage during training.
- Severe **class imbalance**, especially in stroke and Alzheimer datasets.
- Integrating three different disease tasks into a single unified architecture.
- Achieving stable segmentation performance for small lesions.
- Stroke performance plateau due to limited **ISLES Challenge dataset size**.
- Ensuring compatibility of model checkpoints across architecture updates.
- High training time on **Tesla T4 GPU** due to 3D computation.



CONCLUSION

- NeuroX successfully integrates tumor, stroke, and Alzheimer's detection into a single MRI analysis framework
- The proposed system combines 3D CNN, transformer bottleneck, and task-specific decoders for multi-disease prediction
- Accurate lesion segmentation was achieved for tumor and stroke cases
- Uncertainty estimation improved prediction reliability for clinical interpretation
- 3D visualization and automated reporting enhance practical usability
- The system demonstrates strong potential as an AI-assisted neuroimaging tool



- Integrate additional neurological disorders such as haemorrhage and multiple sclerosis
- Expand stroke training using larger datasets beyond ISLES Challenge
- Improve real-time inference speed for clinical deployment
- Add multi-sequence MRI fusion (T1 + T2 + FLAIR) Validate on external hospital datasets
- Explore PACS integration for hospital workflow deployment





THANK YOU

Any Questions?